known as the trace elements. Some micro minerals are iron, iodine, zinc, copper, fluorine, selenium, chromium, manganese, cobalt and molybdenum.

5.3.1 Minerals

The minerals calcium, phosphorus, iron iodine, sodium and others are found in various foods in combination with organic and inorganic compounds. Minerals are necessary for body building, for building bones, teeth and structural parts of soft tissues.

(i) Calcium

- **Distribution:** Calcium makes up between 1.5 to 2 percent of body weight accounting for 1200-1600 g of the adult male body. Ninety percent of calcium is found in mineralized tissues such as bones and teeth as calcium phosphate and calcium carbonate. The remaining 1% is found in blood, extracellular fluid (ECF), muscle and tissues.
- **Functions:** The functions of calcium in humans are manifold:
  - **Bone formation:** The important minerals within bone are calcium phosphate and magnesium. There is 1 kg of calcium in the adult skeleton as a complete crystalline material with phosphate.
  - **Tooth formation:** The enamel and dentin of tooth contain considerable amounts of calcium which are dense and are present along with keratin.
  - **Growth:** It is required for growth as it forms an important part of the bones and teeth and proper functioning of every cell in the body.
  - **Blood clotting:** Calcium contributes to clotting of blood.
  - **Contraction of the muscle:** Calcium ions are bound by the electrostatic forces to the proteins inside and outside the cells and to cell membranes. Proteins bound by calcium alter their configuration at the neuro muscular junction by the nerve impulses causing free calcium to be released. The free calcium bound to troponin leads to an internal trigger and so the contraction of the muscle takes place
  - **Metabolic essentiality:** Calcium acts as a activator for the enzyme renin present in gastric juice which aids the digestion of milk.
- **Food sources:** Calcium is present in both animal and plant foods. The richest source of calcium among animal foods is milk and



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